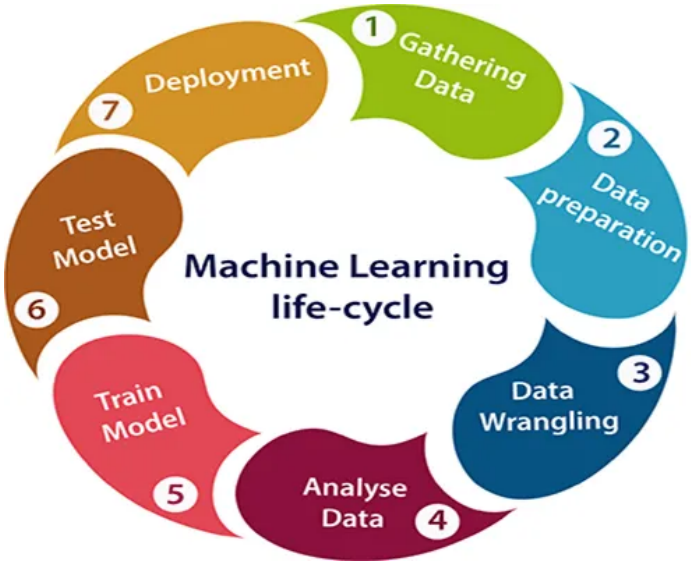
Machine Learning

Machine Learning: Machine Learning is a subset of artificial intelligence that focuses on developing algorithms and statistical models that allow computers to learn and improve their performance on a specific task without being explicitly programmed. It involves training models on data and making predictions or decisions based on that data.

Topic	Traditional Programming	Machine Learning
Definition	Programming is the process of writing code and giving instructions to a computer to perform specific tasks. It involves creating algorithms, designing logic, and implementing solutions using programming languages like Python, Java, C++, etc.	Machine Learning is a subset of artificial intelligence that focuses on developing algorithms and statistical models that allow computers to learn and improve their performance on a specific task without being explicitly programmed. It involves training models on data and making predictions or decisions based on that data.
Development Process	The development process is generally linear and predictable, focusing on implementing and debugging predefined logic.	Involves an iterative process where models are trained, evaluated, and fine-tuned. This process can be less predictable and more experimental.
Data Dependency	Traditional programming generally doesn't depend on large amounts of data. The logic and rules are explicitly provided in the code.	Machine learning algorithms heavily depend on data for training. The quality and quantity of data play a crucial role in the performance of the model.
Adaptability	Traditional programs are typically static and don't adapt or improve on their own. Changes or improvements require manual intervention and reprogramming.	Machine learning models are designed to be adaptive. They can continuously learn from new data and improve their performance over time without the need for manual reprogramming.
Application	Programming is used for a wide range of tasks, including software development, web development, system administration, and more.	Machine learning is applied in various fields like image and speech recognition, natural language processing, recommendation systems, autonomous vehicles, and more.
Skillset	Proficiency in programming languages, algorithms, and data structures is essential for	In addition to programming skills, machine learning requires knowledge of statistics, linear

	programming tasks.	algebra, and understanding of machine learning algorithms and frameworks.
Outcome Predictability	The outcome is highly predictable if the inputs and the logic are known.	Predictions or decisions made by a machine learning model can sometimes be less interpretable, especially with complex models like deep neural networks.

Machine Learning Life Cycle



Machine learning enables computer systems to learn and improve automatically without being explicitly programmed. This process can be described using the machine learning life cycle, which is a series of steps to build an effective machine learning model. The primary goal of this life cycle is to find a solution to a given problem or project.
The machine learning life cycle involves seven major steps:

1. **Gathering Data**
2. **Data Preparation**
3. **Data Wrangling**
4. **Data Analysis**
5. **Training the Model**
6. **Testing the Model**
7. **Deployment**

Before starting the life cycle, it is crucial to understand the problem thoroughly, as a good understanding leads to better results. In this life cycle, a machine learning system called a "**model**" is created by providing "**training**" using data. Therefore, the process begins with data collection.

1. Gathering Data

Data gathering is the first step in the machine learning life cycle. The objective is to identify and obtain all the data relevant to the problem. This involves:

- Identifying various data sources
- Collecting data from these sources (files, databases, the internet, mobile devices, etc.)
- Integrating the collected data
  The quality and quantity of the collected data will significantly impact the efficiency of the model. A coherent set of data, called a dataset, is obtained after performing these tasks, and it is used in subsequent steps.

2. Data Preparation

Once data is collected, it needs to be prepared for further steps. Data preparation involves placing the data in a suitable format and randomizing its order. This step is divided into two processes:

- **Data Exploration:** Understanding the nature of the data, including its characteristics, format, and quality. This helps in identifying correlations, general trends, and outliers.
- **Data Pre-processing**: Preparing the data for analysis by cleaning and transforming it.

3. Data Wrangling

Data wrangling involves cleaning and converting raw data into a usable format. It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:

- Missing Values
- Duplicate data
- Invalid data
- Noise

4. Data Analysis

The cleaned and prepared data is then analyzed. This step involves:

- Selecting analytical techniques
- Building models
- Reviewing results
-

The aim is to build a machine learning model using various analytical techniques and evaluate the outcome. The type of problem is determined, and appropriate machine learning techniques (such as Classification, Regression, Cluster Analysis, Association, etc.) are selected to build and evaluate the model.

5. Training the Model

In this step, the model is trained to improve its performance. The training involves using datasets to help the model understand patterns, rules, and features through various machine learning algorithms.

6. Testing the Model

After training, the model is tested to check its accuracy. A test dataset is provided to the model, and its performance is evaluated to determine the accuracy percentage as per the project requirements.

7. Deployment
The final step in the machine learning life cycle is deployment. If the model meets accuracy and performance requirements, it is deployed in a real-world system. Before deployment, it is ensured that the model improves performance using available data. The deployment phase is akin to making the final project report.

Importance of machine learning:

1. **Automation of Analytical Model Building**: Machine learning automates model construction, enabling systems to learn and improve from data without explicit programming. This saves time and reduces human effort in developing complex models.
2. **Continuous Improvemen**t: Machine learning systems adapt and evolve with new data, leading to better performance and more accurate predictions over time.
3. **Handling Large Volumes of Data**: Machine learning efficiently analyzes vast and complex datasets, quickly extracting meaningful insights that are impossible to obtain manually.
4. **Pattern Recognition:** Machine learning excels at identifying patterns and trends in data, crucial for tasks like fraud detection, image recognition, and market analysis.
5. **Improved Decision Making**: By learning from previous data, machine learning models make reliable and repeatable decisions, enhancing decision-making in finance, healthcare, marketing, and other fields.
6. **Scalability**: Machine learning models can scale to handle increased data and more complex tasks without a proportional increase in resources, essential for real-time processing and analysis.
7. **Applications Across Various Industries:**
    - Healthcare: Diagnosing diseases, personalizing treatment plans, predicting patient outcomes.
    - Finance: Credit scoring, fraud detection, algorithmic trading, risk management.
    - Retail: Personalized recommendations, inventory management, customer sentiment analysis.
    - Manufacturing: Predictive maintenance, quality control, supply chain optimization.
    - Transportation: Autonomous vehicles, route optimization, demand forecasting.
8. **Enhanced Customer Experience**: Machine learning algorithms personalize user experiences by recommending products, content, or services based on individual preferences, leading to higher customer satisfaction.
9. **Cost Efficiency**: Automating tasks and improving efficiency with machine learning can significantly reduce operational costs, such as preventing costly equipment failures through predictive maintenance.

10. **Future-Proofing Businesses:** Adopting machine learning keeps businesses competitive and innovative, allowing quick adaptation to new challenges and opportunities.
11. **Reinforcement Learning**: Reinforcement learning, a type of machine learning, is essential for robotics, gaming, and navigation. It allows systems to learn optimal actions through trial and error, maximizing rewards over time.

Applications of Machine Learning [Audio, Image, Video, Text]

1. **Image Recognition**: Used to identify objects, persons, places in digital images. Example: Facebook's auto friend tagging uses machine learning for face detection and recognition.
2. **Speech Recognition**: Converts voice instructions into text. Used in Google Assistant, Siri, Cortana, and Alexa to follow voice commands.
3. **Traffic Prediction:**Google Maps predicts traffic conditions by using real-time location data and historical traffic patterns.
4. **Product Recommendations**: E-commerce platforms like Amazon and Netflix use machine learning to suggest products and content based on user behavior and interests.
5. **Self-Driving Cars**: Companies like Tesla use machine learning to develop autonomous vehicles that detect people and objects using unsupervised learning methods.
6. **Email Spam and Malware Filtering**: Filters emails into categories such as important, normal, and spam using algorithms like Multi-Layer Perceptron, Decision Tree, and Naïve Bayes.
7. **Virtual Personal Assistants:** Assistants like Google Assistant, Alexa, and Siri use machine learning to process voice commands and perform tasks like playing music, calling, and scheduling.
8. **Online Fraud Detection:** Machine learning secures online transactions by detecting fraudulent activities through pattern recognition and neural networks.
9. **Stock Market Trading:** Predicts stock market trends using long short-term memory neural networks to manage the risk of share price fluctuations.
10. **Medical Diagnosis**: Helps diagnose diseases and predict the exact position of brain lesions, enhancing medical technology for detecting brain tumors and related diseases.
11. **Automatic Language Translation:** Google Neural Machine Translation (GNMT) translates text into familiar languages, facilitating communication in foreign places.

Types of ML

ML can be divided into four main types:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning

Classification of Supervised Machine Learning

Supervised machine learning involves training algorithms on labeled datasets, where the input data is paired with the correct output. It can be further classified into two categories:

a. **Classification**:
   i. **Random Forest Algorithm**: Combines multiple decision trees to improve accuracy and control over-fitting.
   ii. **Decision Tree Algorithm**: Splits data into branches to predict outcomes based on input variables.
   iii. **Logistic Regression Algorithm**: Estimates the probability of a binary outcome based on one or more predictor variables.
   iv. **Support Vector Machine Algorithm**: Finds the hyperplane that best separates different classes in the input data.
b. **Regression**:
   i. **Simple Linear Regression Algorithm**: Models the relationship between two continuous variables by fitting a linear equation.
   ii. **Multivariate Regression Algorithm**: Extends linear regression to model the relationship between multiple input variables and a continuous output variable.
   iii. **Decision Tree Algorithm**: Also used for regression tasks to predict continuous outputs by splitting data into decision branches.
   iv. **Lasso Regression**: Adds a penalty to the regression model to shrink coefficients of less important features, preventing overfitting.

Classification of Unsupervised Machine Learning

Unsupervised learning algorithms work with unlabeled data, finding patterns and structures without explicit instructions. It can be divided into two main categories:

a. **Clustering**:
   i. **K-Means Clustering Algorithm:** Partitions data into K clusters based on feature similarity.
   ii. **Mean-Shift Algorithm**: Clusters data by shifting points towards the mode (highest density area) iteratively.
   iii. **DBSCAN Algorithm**: Identifies clusters of varying shapes based on density, making it robust to noise.
   iv. **Principal Component Analysis (PCA):** Reduces dimensionality by transforming data to a new coordinate system.
   v. **Independent Component Analysis (ICA):** Separates a multivariate signal into additive, independent components.
b. **Association**:
   i. **Apriori Algorithm:** Finds frequent itemsets and generates association rules for market basket analysis.
   ii. **Eclat Algorithm:** Efficiently discovers frequent itemsets using a depth-first search strategy.
   iii. **FP-Growth Algorithm**: Uses a frequent pattern tree to compress data and find frequent itemsets without candidate generation.

Reinforcement Learning
Reinforcement learning (RL) is a feedback-based machine learning technique where an agent learns to make decisions by performing actions and receiving rewards or penalties. The goal is to maximize cumulative rewards over time. Unlike supervised learning, RL does not require labeled data and learns through exploration and exploitation of its environment.
Examples of Reinforcement Learning:

- **Video Games:** AlphaGo and AlphaGo Zero: Developed by DeepMind, these RL algorithms learned to play the game of Go at a super-human level through self-play and reinforcement learning.
- **Resource Management:** Automatic Resource Scheduling: RL algorithms are used in cloud computing to optimize the allocation of resources, minimizing job waiting times and improving efficiency.
- **Robotics**: Intelligent Industrial Robots: RL is applied in robotics to enable robots to learn tasks such as assembly, navigation, and manipulation in dynamic environments.
- **Text Mining**: NLP Applications by Salesforce: Reinforcement learning enhances natural language processing tasks such as text summarization and translation by learning from large datasets and improving over time.

Advantages of Reinforcement Learning:

- Solves complex problems that are difficult for traditional methods.
- Mimics human learning, leading to more accurate results.
- Achieves long-term goals through cumulative rewards.

Disadvantages of Reinforcement Learning:

- Not suitable for simple problems due to high complexity.
- Requires extensive data and computational resources.
- Overload of states can weaken results due to the curse of dimensionality.

Common Terminologies

Model

In general, any mathematical construct that processes input data and returns output. A model is the set of parameters and structure needed for a system to make predictions. In supervised machine learning, a model takes an example as input and infers a prediction as output. Within supervised machine learning, models differ somewhat. For example:

- A linear regression model consists of a set of weights and a bias.
- A neural network model consists of:
    - A set of hidden layers, each containing one or more neurons.
    - The weights and bias associated with each neuron.
- A decision tree model consists of:
    - The shape of the tree; that is, the pattern in which the conditions and leaves are connected.
    - The conditions and leaves.

Unsupervised machine learning also generates models, typically a function that can map an input example to the most appropriate **cluster**.

Features

Features are individual measurable properties or characteristics of a phenomenon being observed. In machine learning, features are the input variables used to train a model. They play a crucial role in predictive modeling, as the quality and relevance of features directly impact the model's performance. Features can be numerical, categorical, text-based, or derived from raw data through feature engineering techniques. Effective feature selection and engineering are key steps in building robust and accurate machine learning models.

Labels

Labels are the output or target variables that the model is trained to predict. In supervised learning, labels are the known outcomes associated with the input features in the training data. For example, in a dataset used for image classification, each image might have a label indicating the object it contains (e.g., "cat" or "dog"). Labels are essential for guiding the learning process, allowing the model to learn the relationship between the input features and the output predictions.

Recall

Recall, also known as sensitivity or true positive rate, is a metric used to evaluate the performance of a classification model. It measures the proportion of actual positive cases that are correctly identified by the model. Recall is calculated as the number of true positives divided by the sum of true positives and false negatives:

$$Recall = True\ Positive\ /\ (True\ Positives + False\ Negatives)$$

A high recall indicates that the model is good at identifying positive cases, which is crucial in scenarios where missing positive cases is costly, such as in medical diagnoses or fraud detection.

Sensitivity

Same as Recall

ROC Curve

[A graph of true positive rate versus false positive rate for different classification thresholds in binary classification.]

The **ROC (Receiver Operating Characteristic)** curve is a graphical representation used to evaluate the performance of a binary classification model. It plots the true positive rate (sensitivity) against the false positive rate (1-specificity) at various threshold settings. Each point on the ROC curve represents a different threshold, showing the trade-off between sensitivity and specificity. A model with a curve closer to the top-left corner indicates better performance. The area under the curve (AUC) quantifies the overall ability of the model to discriminate between positive and negative classes.

AUC Curve

[A number between 0.0 and 1.0 representing a binary classification model's ability to separate positive classes from negative classes. The closer the AUC is to 1.0, the better the model's ability to separate classes from each other.]
The AUC (Area Under the Curve) is a metric used to measure the performance of a binary classification model. Specifically, it represents the area under the ROC (Receiver Operating Characteristic) curve. The AUC value ranges from 0 to 1, with a higher value indicating better model performance. An AUC of 0.5 suggests no discrimination (random guessing), while an AUC of 1.0 indicates perfect discrimination. It provides a single scalar value to compare the effectiveness of different models in distinguishing between positive and negative classes.

Attention

A mechanism used in a neural network that indicates the importance of a particular word or part of a word. Attention compresses the amount of information a model needs to predict the next token/word. A typical attention mechanism might consist of a weighted sum over a set of inputs, where the weight for each input is computed by another part of the neural network.

Agent

An agent is software that autonomously plans and executes a series of actions in pursuit of a goal, with the ability to adapt to changes in its environment. For example, LLM-based agents might use the LLM to generate a plan, rather than applying a reinforcement learning policy.

Centroid

The center of a cluster as determined by a k-means or k-median algorithm. For instance, if k is 3, then the k-means or k-median algorithm finds 3 centroids.

Convergence

A state reached when loss values change very little or not at all with each iteration.

Distillation

The process of reducing the size of one model (known as the teacher) into a smaller model (known as the student) that emulates the original model's predictions as faithfully as possible.

Epoch

A full training pass over the entire training set such that each example has been processed once.

Pooling

Reducing a matrix (or matrixes) created by an earlier convolutional layer to a smaller matrix. Pooling usually involves taking either the maximum or average value across the pooled area.

Intercept

Intercept is a constant value which tells us at what point in the x-y coordinate graph, the regression line must start if it follows a linear regression. Since it is a constant value, hence it is not dependent on any change in independent variables.

Clustering

A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group.

1. **Partitioning Clustering:**
    a. **Definition**: Partitioning methods partition data objects into several non-overlapping subsets or clusters.
    b. **Example Algorithm**: K-Means is a popular partitioning clustering algorithm. It iteratively assigns data points to K clusters based on the nearest mean and updates the cluster centroids until convergence.
2. **Density-Based Clustering:**
    a. **Definition**: Density-based methods partition the data space into regions of varying density. Clusters are formed where the density of data points exceeds a threshold.
    b. **Example Algorithm:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm. It groups together closely packed points based on density, identifying noise as outliers.
3. **Distribution Model-Based Clustering:**
    a. **Definition**: Distribution model-based methods assume that the data is generated from a mixture of probability distributions.
    b. **Example Algorithm**: Gaussian Mixture Models (GMM) is a widely used distribution model-based clustering algorithm. It assumes that the data is generated from a mixture of several Gaussian distributions and assigns probabilities to each point belonging to each cluster.
4. **Hierarchical Clustering:**
    a. **Definition**: Hierarchical clustering builds a hierarchy of clusters. It doesn't require the number of clusters to be specified in advance and can be visualized as a dendrogram.
    b. **Example Algorithm**: Agglomerative Hierarchical Clustering is a common approach where each data point initially represents a single-cluster, and then at each step, the two closest clusters are merged until only one cluster remains.
5. **Fuzzy Clustering:**
    a. **Definition**: Fuzzy clustering allows data points to belong to multiple clusters with varying degrees of membership. Instead of hard assignments, it assigns probabilities of data points belonging to each cluster.
    b. **Example Algorithm**: Fuzzy C-Means is a well-known fuzzy clustering algorithm. It extends K-Means by allowing data points to belong to multiple clusters with different degrees of membership.

Differences between agglomerative and divisive clustering

- **Starting Point**:
  - **Agglomerative Clustering**: Starts with each data point as its own cluster.
  - **Divisive Clustering**: Starts with all data points in a single cluster.
- **Approach**:
  - **Agglomerative Clustering**: "Bottom-Up" approach. Iteratively merges the most similar pairs of clusters.
  - **Divisive Clustering**: "Top-Down" approach. Recursively splits the clusters into smaller sub-clusters.
- **Process**:
  - **Agglomerative Clustering**: Merges clusters based on their similarity until all data points belong to a single cluster.
  - **Divisive Clustering**: Splits a large cluster into smaller sub-clusters based on their dissimilarity.
- **Hierarchy Creation:**
  - **Agglomerative Clustering:** Builds a hierarchy by merging clusters.
  - **Divisive Clustering**: Builds a hierarchy by dividing clusters.
- **Initial State:**
  - **Agglomerative Clustering**: Each data point is initially considered as a separate cluster.
  - **Divisive Clustering**: All data points are initially considered as a single cluster.
- **Final State:**
  - **Agglomerative Clustering**: Ends with a single cluster containing all data points.
  - **Divisive Clustering**: Ends with each data point as its own cluster or predefined number of clusters.
- **Alternate Name:**
  - **Agglomerative Clustering**: Sometimes referred to as "bottom-up" clustering.
  - **Divisive Clustering**: Also known as "divide and conquer" clustering.

Measures of Distance

1. **Minkowski distance**
   a. Zero Vector- Zero vector will have zero length.
   b. Scalar Factor- The direction of the vector doesn't change when you multiply it with a positive number though its length will be changed.
   c. Triangle Inequality- If the distance is a norm then the calculated distance between two points will always be a straight line.
   d. $$\left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{1/p}$$
2. **Manhattan distance**

a. We use Manhattan distance, also known as city block distance, or taxicab geometry if we need to calculate the distance between two data points in a grid-like path.

b. $$d = \sum_{i=1}^{n} |x_i - y_i|$$

3. **Euclidean distance**

a. $$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

4. **Hamming distance**
    a. Hamming distance is a metric for **comparing two binary data strings**. While comparing two binary strings of equal length, Hamming distance is the number of bit positions in which the two bits are different.
    b. In order to calculate the Hamming distance between two strings, and, we perform their **XOR operation, (a⊕ b),** and then count the total number of 1s in the resultant string.
5. **Cosine distance**
    a. Cosine distance & Cosine Similarity metric is mainly used to find similarities between two data points. As the cosine distance between the data points increases, the cosine similarity, or the amount of similarity decreases, and vice versa.
    b. Thus, Points closer to each other are more similar than points that are far away from each other. Cosine similarity is given by **Cos θ**, and cosine distance is **1-Cos θ**.

K Means Clustering Algorithm

K-means clustering, also known as Lloyd's algorithm, is an unsupervised machine learning algorithm that groups similar objects into clusters based on their similarities. The algorithm is iterative and data-partitioning, assigning each observation to one of ***k*** clusters defined by centroids.

1. Select the number K to decide the number of clusters.
2. Select random K points or centroids. (It can be different from the input dataset).
3. Assign each data point to their closest centroid, which will form the predefined K clusters.
4. Calculate the variance and place a new centroid of each cluster.
5. Repeat the third steps, which means re-assign each datapoint to the new closest centroid of each cluster.
6. If any reassignment occurs, then go to step-4 else go to FINISH.

Elbow Method

- It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).
- For each value of K, calculate the WCSS (**Within Cluster Sum of Squares**) value.
- Plots a curve between calculated WCSS values and the number of clusters K.
- The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.

$$\mathbf{WCSS} = \sum_{\text{Pi in Cluster1}} \mathbf{distance}(P_i\ C_1)^2 + \sum_{\text{Pi in Cluster2}} \mathbf{distance}(P_i\ C_2)^2 + \sum_{\text{Pi in CLuster3}} \mathbf{distance}(P_i\ C_3)^2$$

KNN Classification Algorithm

1. Choose the Value of K
2. Calculate Distance:
    a. For a given test data point, calculate the distance between this point and all the training data points. The most common distance metric used is Euclidean distance.
3. Sort Distances
    a. Rank the distances from the smallest to the largest.
4. Select Neighbors
5. Choose the K data points that have the smallest distances to the test point.
6. Voting for Classification (or Averaging for Regression):
    a. **For classification**: The K selected neighbors vote on the class, and the class with the most votes is assigned to the test point.
    b. **For regression**: Calculate the average value of the K nearest neighbors and assign this average to the test point.
7. Predict the Output

How to choose the value of k for KNN Algorithm?

The value of k is very crucial in the KNN algorithm to define the number of neighbors in the algorithm. The value of k in the k-nearest neighbors (k-NN) algorithm should be chosen based on the input data. If the input data has more outliers or noise, a higher value of k would be better. It is recommended to choose an **odd value for k** to avoid ties in classification. **Cross-validation and Square Root methods** can help in selecting the best k value for the given dataset.

- **Square Root Method**: We can consider the square root of all data points in the training dataset as the optimal value of K.
- **Cross-Validation Method**: We will start with K=1, run cross-validation (5 to 10-fold), measure the accuracy, and keep repeating until the results get consistent.

Differences between KNN and K-Means

Topic	KNN	K-Means
Type	**Supervised** learning algorithm	**Unsupervised** learning algorithm
Goal	**Predict class** labels or **regression values** based on labeled training data	**Partition data** into K **clusters** to find patterns/structures
Output	**Predicted class** or **value** for a query point	**Set of K clusters** and their centers
Input	Requires **labeled** data	**Unlabeled** data
Parameters	**Number of neighbors** (K), distance metric	**Number of clusters** (K), initialization of cluster centers
Complexity	**Lazy algorithm**; computationally intensive during **prediction**	**Eager algorithm**; computationally intensive during **training**
Use Cases	**Small**, clean datasets needing classification or regression	**Large**, unlabeled datasets needing pattern recognition or **segmentation**
Pros	Simple, flexible, handles non-linear data	Fast, scalable, reduces dimensionality and noise
Cons	**Slow for large datasets**, memory-intensive, affected by the curse of dimensionality	**Sensitive** to initial center selection, outliers, and cluster shape

Linear Regression [Simple/Multiple]

Linear regression is a machine learning algorithm which estimates how a model is following a linear relationship between one response variable (denoted by y) and one or more explanatory variables (denoted by X1, X2, X3…. Xn). The response variable will depend on how the explanatory variables change and not the other way round. Response variable is also known as target or dependent variable while the explanatory variable is known as independent or predictor variables.

- **Simple Linear Regression**: It is a type of linear regression model where there is only an independent or explanatory variable.
- **Multiple Linear Regression**: It is similar to simple linear regression but here we have more than one independent or explanatory variable.

Steps

1. Define the Variables:
    a. y is the vector of dependent variable values.
    b. X is the matrix of input features (with a column of ones for the intercept).
2. Construct the Matrices:

a. $$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

3. Calculate the Coefficients
    a. Using the normal equation, calculate the coefficients β
    b. $$\beta = (X^T X)^{-1} X^T y$$
    c. Here, β is a vector of the coefficients (including the intercept).
4. Interpret the Coefficients:
    a. The first element of β is the intercept (β_0).
5. Make Predictions:
    a. $\mathbf{y_{new} = X_{new}\beta}$

Logistic Regression

Log of Odds

- In logistic regression, the log-odds are modeled as a linear combination of predictor variables.
- This transformation helps in handling the probabilities directly in a regression framework.
- The log-odds are particularly useful because they can take any real number value, unlike probabilities which are bounded between 0 and 1.
- 3 Types:
    - **Binomial**: In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
    - **Multinomial**: In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep"
    - **Ordinal**: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".

Steps

1. Formulate the Sigmoid Function:
    a. $S(z) = 1/(1+e^{-z})$; ***z = log of odds***
    b. The sigmoid function is a mathematical function used to map the predicted values to probabilities.
    c. It maps any real value into another value within a range of 0 and 1.
    d. The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
2. $$log\left[\frac{y}{1-y}\right] = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n$$

Differences between Linear Regression & Logistic Regression

- **Type of prediction**
  - Linear regression predicts continuous values, while logistic regression predicts binary or categorical outcomes
- **Regression line**
  - Linear regression has a straight line, while logistic regression has an S-shaped curve
- **Equation model**
  - Linear regression uses an ordinary least squares method, while logistic regression uses a complex equation model
- **Mathematical distribution**
  - Linear regression follows a normal distribution, while logistic regression follows a binomial distribution
- **Computing speed**
  - Linear regression is faster than logistic regression
- **Uses**
  - Linear regression is used for regression problems, while logistic regression is used for classification problems
- **Assumptions**
  - Linear regression assumes constant variance of errors, while logistic regression does not

Support Vector Machine (SVM)

A supervised machine learning algorithm used for both classification and regression. it's best suited for classification. The main objective of the SVM algorithm is to find the optimal hyperplane in an N-dimensional space that can separate the data points in different classes in the feature space. The hyperplane tries that the margin between the closest points of different classes should be as maximum as possible. The dimension of the hyperplane depends upon the number of features. If the number of input features is two, then the hyperplane is just a line. If the number of input features is three, then the hyperplane becomes a 2-D plane.

- Kernel
  - A function that transforms the data into a higher-dimensional space to make it easier to find a separating hyperplane. Common kernels include linear, polynomial, radial basis function (RBF), and sigmoid.
- Soft Margin
  - A concept that allows some misclassification of data points to achieve better generalization on unseen data. It introduces a penalty for misclassified points.
- Margin
  - The distance between the hyperplane and the nearest data point from either class. SVM aims to maximize this margin.
- Hyperplane
  - Hyperplane is the decision boundary that is used to separate the data points of different classes in a feature space.

Naïve Bayes classifier

The Naïve Bayes classifier is a supervised machine learning algorithm that is used for classification tasks such as text classification. They use principles of probability to perform classification tasks.

- Prepare the Dataset
  - Collect the dataset and divide it into training and test sets. Each data point in the dataset consists of features X = (X_1 , X_2, X_3...X_n) & a corresponding class label **Y**.
- Calculate **Prior Probabilities**
  - Calculate the prior probability for each class y
    - $$P(Y = y) = \frac{\text{Number of instances of class } y}{\text{Total number of instances}}$$
- Calculate **Likelihood**
  - Calculate the Tables for each P(X=x | Y = y)
- Calculate **Posterior Probabilities**
  - $$P(Y = y|X_{\text{new}}) = \frac{P(Y = y)\prod_{i=1}^{n} P(X_i = x_i|Y = y)}{P(X_{\text{new}})}$$

Decision Tree

A decision tree is a supervised machine learning algorithm that uses input data to model and predict outcomes. It's a tree-like structure with different types of nodes, including root, internal, and leaf nodes:

- Internal nodes: Test on an attribute
- Leaf nodes: Represent the final decision or prediction
- Steps in ID3 algorithm:
  - It begins with the original set S as the root node.
  - On each iteration of the algorithm, it iterates through the very unused attribute of the set S and calculates Entropy(H) and Information gain(IG) of this attribute.
  - $$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i \qquad E(T, X) = \sum_{c \in X} P(c)E(c)$$
  - It then selects the attribute which has the **Largest Information gain**.
  - The set S is then split by the selected attribute to produce a subset of the data.
  - The algorithm continues to recur on each subset, considering only attributes never selected before.
- Steps in Gini Impurity algorithm:
  - $$Gini = 1 - \sum_{i=1}^{n} p_i^2$$
  - IG: **Gini(child1) x P_{child1} + Gini(child2) x P_{child2}**
  - It then selects the attribute which has the **Smallest Information gain**.

EXTRA

CART (Classification And Regression Tree)

CART is a predictive algorithm used in Machine learning and it explains how the target variable's values can be predicted based on other matters. It is a decision tree where each fork is split into a predictor variable and each node has a prediction for the target variable at the end. The term CART serves as a generic term for the following categories of decision trees:

- Classification Trees: The tree is used to determine which "class" the target variable is most likely to fall into when it is continuous.
- Regression trees: These are used to predict a continuous variable's value.

In the decision tree, nodes are split into sub-nodes based on a threshold value of an attribute. The root node is taken as the training set and is split into two by considering the best attribute and threshold value. Further, the subsets are also split using the same logic. This continues till the last pure sub-set is found in the tree or the maximum number of leaves possible in that growing tree.

Federated Learning

Federated learning is a distributed machine learning paradigm that enables machine learning models to be trained on decentralized data sources without compromising privacy or security. In traditional machine learning, the data is centralized, meaning that the data is collected, stored, and processed in a single location. This data centralization can create privacy and security issues, as sensitive information is often collected, stored, and processed in this manner.

In contrast, federated learning enables machine learning models to be trained on decentralized data sources by allowing multiple parties to train models on their local data while sharing the learned parameters with the central server. This allows the server to aggregate the models and update the global model while preserving the privacy and security of the data.

- **Advantages**:
    - Privacy and Security: Data remains decentralized, protecting sensitive information such as medical or financial records.
    - Scalability: Easily accommodates large datasets and updates the global model as data sources grow.
    - Cost-effectiveness: Reduces the need for large data centers and associated costs.
    - Decentralization: Allows distributed data sources, enabling collaboration across different locations.
    - Improved Performance: Enhances model performance by training on diverse datasets.
- **Limitations**:
    - Latency: Updating models can be slow, affecting performance and accuracy.
    - Bandwidth: Requires high-bandwidth networks for effective model transmission between data sources and the central server.

    - Trust: Depends on trust between data sources and the central server, as untrustworthy sources can compromise the global model.
    - Technical Expertise: Implementation requires advanced technical skills for developing complex algorithms and protocols.

Incremental Learning

Incremental learning is a machine learning technique that continuously updates a model as new data becomes available. It's also known as continual learning or lifelong learning. Incremental learning is useful when training data is available gradually or when the data is too large for the system's memory. It's especially important in scenarios with streaming data, where the model needs to adapt quickly. For example, a smart thermostat that learns a user's temperature preferences is an incremental learning problem.

- Steps:
    - Data collection and preprocessing
    - Model initialization
    - Incremental updates
    - Spaced repetition
    - Model evaluation
    - Concept drift detection
    - Regularization and forgetting
    - Batching and mini-batching
- Challenges
    - Catastrophic forgetting: The model may forget old information as it learns new data
    - Concept drift: It can be difficult to handle abrupt changes in data trends
    - Overfitting: The model may over-adjust its parameters based on recent data, which may not represent the overall distribution

Transfer Learning

Transfer learning (TL) is a machine learning (ML) technique that reuses knowledge from a previous task to improve performance on a related task. For example, knowledge gained from learning to recognize cars could be used to recognize trucks. TL involves fine-tuning a model that has been pre-trained on one task for a new, related task. Training a new ML model can be time-consuming and requires a lot of data, computing power, and iterations before it's ready for production.

Generative AI

Generative AI refers to algorithms that can create new data instances that resemble a given dataset. This can include generating text, images, music, and more. Generative AI relies on sophisticated machine learning models called deep learning models—algorithms that simulate the learning and decision-making processes of the human brain.

- Three phases:
    - **Training**, to create a foundation model that can serve as the basis of multiple gen AI applications.
    - **Tuning**, to tailor the foundation model to a specific gen AI application.
    - **Generation**, **evaluation and returning**, to assess the gen AI application's output and continually improve its quality and accuracy.
- Key Techniques:
    - Generative Adversarial Networks (GANs): Consist of two neural networks, a generator and a discriminator, which compete to produce realistic data.
    - Variational Autoencoders (VAEs): Use probabilistic graphical models to encode input data into a latent space and decode it back to data.
    - Transformers: Particularly used in natural language processing, they generate text by predicting subsequent words in a sequence.

Explainable AI

Explainable AI (XAI) is a collection of methods and processes that help people understand and trust the results of machine learning algorithms. It can also refer to an artificial intelligence system that allows humans to maintain intellectual oversight.

- Transparency: Explainable AI (XAI) aims to make AI decision-making processes clear and understandable to humans.
- Interpretability: XAI techniques ensure AI outputs are easy to interpret and explain in human terms.
- Accountability: By explaining AI decisions, XAI enhances accountability and ensures compliance with ethical and legal standards.
- Trust and Adoption: Explainability builds trust in AI systems, essential for their adoption in critical fields.
- Debugging and Improvement: Understanding AI decisions helps identify and correct errors, improving model reliability.

Transformers

Transformers are a type of deep learning model that use self-attention to process sequential input data for natural language processing (NLP) and computer vision (CV) tasks. They are made up of an encoder and a decoder that work together to process input and generate output.

- Self-Attention Mechanism: Transformers use self-attention to weigh the importance of different words in a sentence, allowing the model to focus on relevant parts of the input.
- Parallelization: Unlike RNNs, transformers process input data in parallel, leading to faster training times and improved efficiency.
- Scalability: Transformers are highly scalable, capable of handling large datasets and complex tasks, making them suitable for NLP models like BERT and GPT.
- Pretraining and Fine-Tuning: Transformers benefit from a two-step process of pretraining on large datasets and fine-tuning on specific tasks, enhancing their performance and versatility.
- Versatility: Beyond NLP, transformers are being adapted for various applications, including image processing, protein folding, and more.

BERT

BERT (Bidirectional Encoder Representations from Transformers) is an advanced natural language processing (NLP) model developed by Google. Unlike traditional models that read text in a single direction, BERT processes text bidirectionally, understanding context from both directions. This approach allows BERT to capture the nuanced meaning of words in relation to their surrounding words.

- Pre-training: BERT is pre-trained on large text corpora using two tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In MLM, random words in sentences are masked and the model learns to predict them. In NSP, the model predicts if a given sentence logically follows another.
- Fine-tuning: After pre-training, BERT is fine-tuned on specific tasks (e.g., sentiment analysis, question answering) with labeled data. This customization helps BERT perform well across various NLP tasks.
- Architecture: BERT is based on transformers, specifically using multiple layers of bidirectional transformers to encode the input text. This deep and layered structure allows BERT to understand complex linguistic patterns and relationships.

AutoML

Automated machine learning (AutoML) is the process of automating the tasks of applying machine learning to real-world problems. AutoML potentially includes every stage from beginning with a raw dataset to building a machine learning model ready for deployment.

- Data Preprocessing: AutoML automates data cleaning, transformation, and feature engineering, ensuring that data is in the optimal format for modeling.
- Model Selection: It automates the selection of the best algorithm from a pool of candidates, considering the specific characteristics of the dataset.
- Hyperparameter Tuning: AutoML optimizes the hyperparameters of the chosen model to improve performance.
- Model Training and Evaluation: It trains multiple models, evaluates their performance using cross-validation or other methods, and selects the best-performing model.
- Deployment: AutoML streamlines the process of deploying the model into production environments.

AI vs ML vs DL vs DS

- Artificial Intelligence (AI)
    - Definition: Simulation of human intelligence in machines.
    - Techniques: Rule-based systems, expert systems, natural language processing.
    - Applications: Robotics, autonomous vehicles, game playing.
    - Complexity: Broadest, includes ML and DL.
- Machine Learning (ML)
    - Definition: Algorithms that learn from data.
    - Techniques: Supervised learning, unsupervised learning, reinforcement learning.
    - Applications: Spam detection, recommendation systems, fraud detection.
    - Complexity: Subset of AI.

- Deep Learning (DL)
    - Definition: Neural networks with many layers.
    - Techniques: Convolutional neural networks, recurrent neural networks.
    - Applications: Image recognition, speech recognition, natural language processing.
    - Complexity: Subset of ML, focuses on large datasets and complex patterns.
- Data Science (DS)
    - Definition: Extraction of insights from data.
    - Techniques: Statistical analysis, data mining, ML.
    - Applications: Business analytics, predictive modeling, data visualization.
    - Complexity: Interdisciplinary, includes aspects of AI, ML, and DL.

Tensorflow vs Keras vs PyTorch

- TensorFlow
    - Developed By: Google Brain
    - Language: Primarily Python, but supports C++, JavaScript, and others
    - Features:
        - Comprehensive ecosystem with tools like TensorBoard and TensorFlow Lite.
        - High performance with distributed computing support.
        - Suitable for production deployment.
    - Usage: Widely used for both research and production in industries.
- Keras
    - Developed By: Initially an independent project by François Chollet, now part of TensorFlow
    - Language: Python
    - Features:
        - High-level API designed for ease of use and rapid development.
        - Integrates seamlessly with TensorFlow.
        - Simplifies building and training neural networks.
    - Usage: Preferred for quick prototyping and beginners.
- PyTorch
    - Developed By: Facebook's AI Research lab (FAIR)
    - Language: Python, with support for C++ backend
    - Features:
        - Dynamic computation graph for easy debugging and flexibility.
        - Strong support for GPU acceleration.
        - Popular in academic research.
    - Usage: Favored by researchers and those needing dynamic computation graphs.

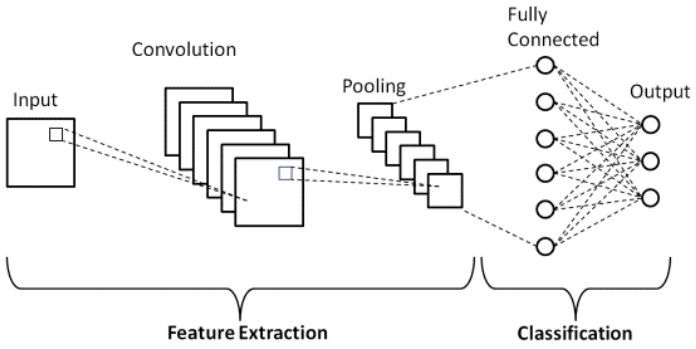
Automated Feature Reduction

Automated feature reduction, also known as feature selection or dimensionality reduction, is a critical aspect of machine learning model development. It involves selecting a subset of relevant features (input variables) from the original set of features to improve model performance, reduce

computational complexity, and enhance interpretability. Here are some common techniques for automated feature reduction:

- Filter Methods
- Wrapper Methods
- Embedded Methods.
- Dimensionality Reduction Techniques.
- Autoencoders
- Genetic Algorithms
- Feature Importance from Tree-based Models

CNN



CNN stands for Convolutional Neural Network, a type of deep neural network primarily used for image recognition and processing, although it's also applied to various other tasks like natural language processing and speech recognition.

- **Convolutional Layers:** These layers apply convolution operations to the input data. Convolution involves sliding a filter (also called a kernel) over the input data and computing dot products to produce feature maps. This operation captures spatial hierarchies of features in the data.
- **Pooling Layers**: Pooling layers downsample the feature maps generated by convolutional layers, reducing their dimensionality. Max pooling, for instance, selects the maximum value from each patch of the feature map, effectively preserving the most salient features.
- **Activation Functions:** Activation functions introduce non-linearity into the network, allowing it to learn complex patterns and relationships in the data. Common choices include **ReLU (Rectified Linear Unit)**, which replaces negative values with zero, and its variants.
- **Training**: CNNs are trained using backpropagation and optimization algorithms like stochastic gradient descent (SGD) or its variants.
- **Transfer Learning**: CNNs often leverage transfer learning, where a pre-trained model is fine-tuned on a new dataset/task.

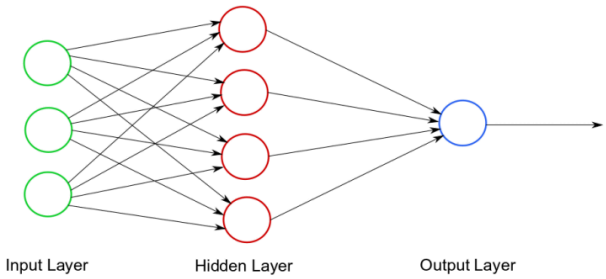
CNNs have revolutionized various fields by achieving state-of-the-art performance in tasks like image classification, object detection, and segmentation. Their **hierarchical architecture** enables them to automatically learn hierarchical features from raw data, making them powerful tools in machine learning and computer vision.

Neural Network

Neural networks (NNs) are a class of machine learning algorithms inspired by the structure and function of the human brain. They consist of interconnected nodes, or artificial neurons, organized into layers. Here's a brief overview:

- Input Layer: The input layer receives the initial data and passes it on to the next layer
- Hidden Layers: Hidden layers are intermediary layers between the input and output layers. Each neuron in a hidden layer receives inputs from the previous layer, performs a weighted sum of these inputs, adds a bias term, and applies an activation function to produce an output.
- Output Layer: The output layer produces the final result or prediction. The number of neurons in this layer depends on the nature of the task.
- Weights and Biases: Neural networks learn from data by adjusting parameters called weights and biases. The weights determine the strength of connections between neurons, while biases control the neuron's output threshold.
- Activation Functions: Activation functions introduce non-linearity into the network, allowing it to learn complex patterns in the data.
- Training: **Neural networks are trained using labeled data**.

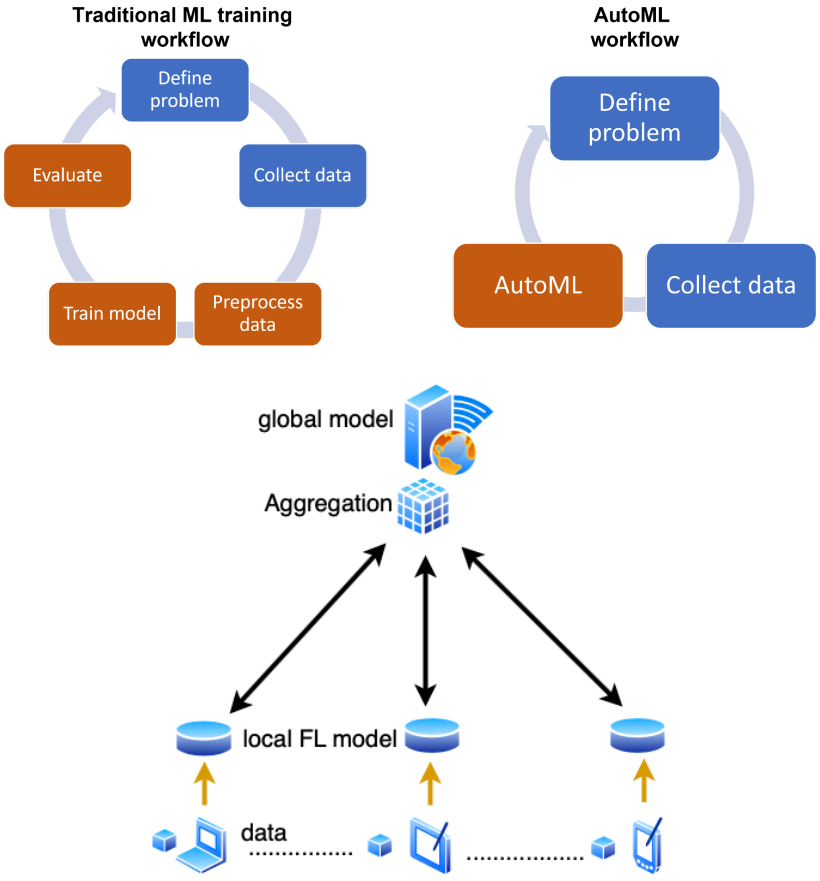
Neural networks have demonstrated remarkable capabilities in a wide range of applications, including image and speech recognition, natural language processing, and autonomous vehicles.



Backpropagation

Backpropagation is a mathematical algorithm used in machine learning and artificial intelligence to train artificial neural networks. It's a type of supervised learning that uses a known output for each input value to calculate the loss function gradient. The algorithm works by feeding error rates backward through the neural network layers to fine-tune the weights. It is used in:

- Training feedforward neural networks, such as convolutional neural networks
- Predictive analytics, machine learning applications that involve predictive analytics


Traditional ML training workflow
Define problem
Collect data
Preprocess data
Train model
Evaluate
AutoML workflow
Define problem
Collect data
AutoML
global model
Aggregation
local FL model
data